

EENG250 ELECTRIC CIRCUITS I · CHAPTER 3

Simple Resistive Circuits

Combine resistors, split a voltage or a current in one line, and untangle a bridge with a delta-to-wye step.

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PREDICT FIRST

A resistive touch screen finds your finger with one voltage measurement per direction. How?

TODAY

By the end of this chapter you can

1 • Series and parallel

Recognise them, and reduce a network to one equivalent resistance.

2 • Division

Split a voltage across series resistors and a current between parallel ones.

3 • Loading

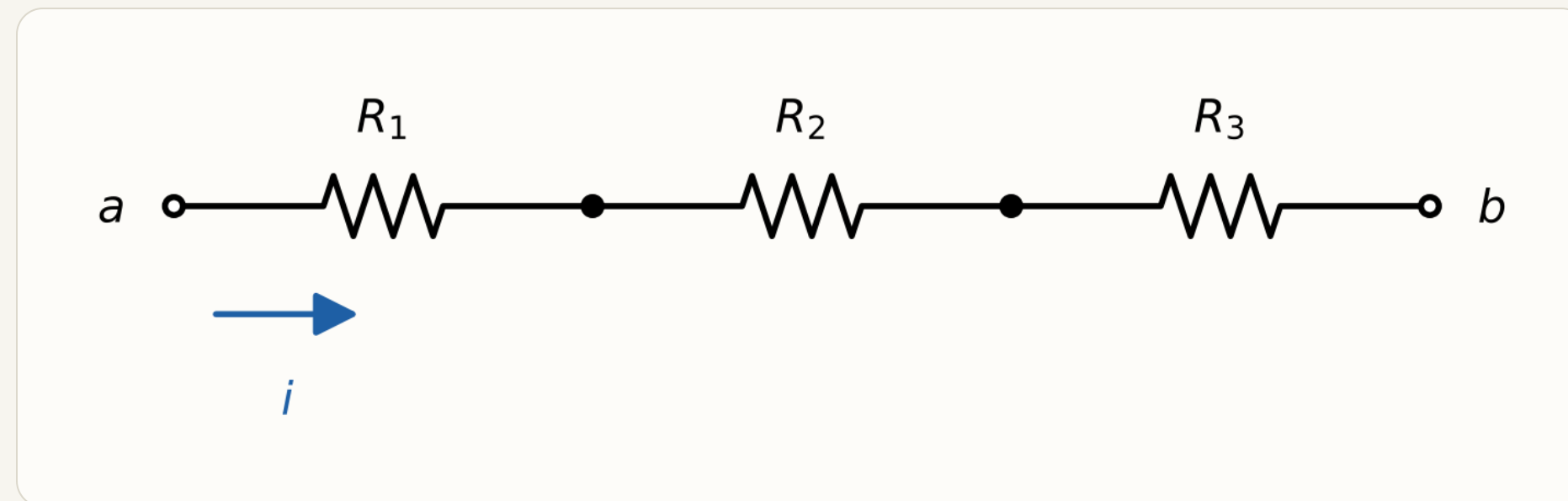
Design a voltage divider and predict what a load does to it.

4 • Delta and wye

Convert Δ to Y and Y to Δ to solve a bridge.

RESISTORS IN SERIES

Series resistors carry the same current, so their resistances add

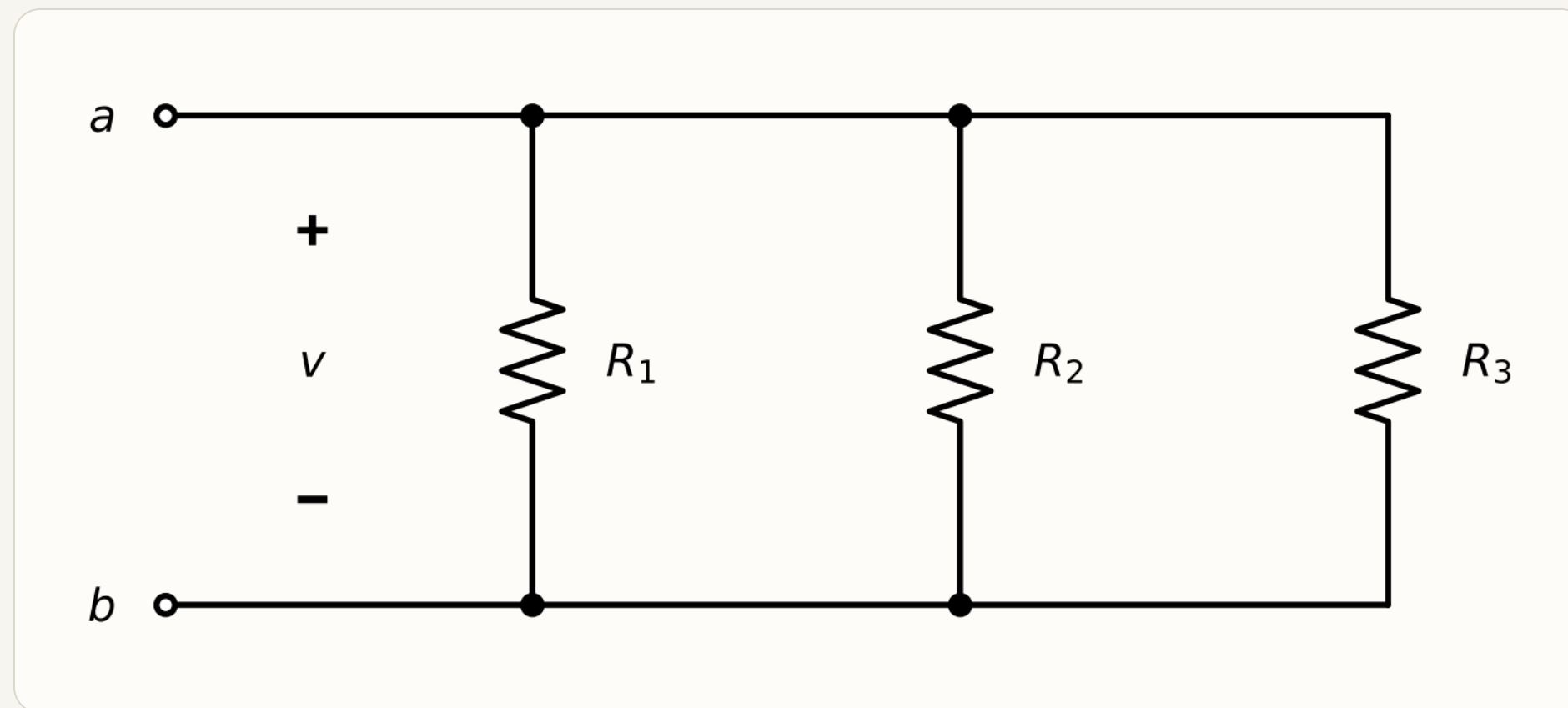


$$R_{eq} = R_1 + R_2 + R_3 + \dots$$

Two elements are in series when they are the **only** elements at the node between them: KCL then forces the same current through both.

RESISTORS IN PARALLEL

Parallel resistors share two nodes, so they share one voltage



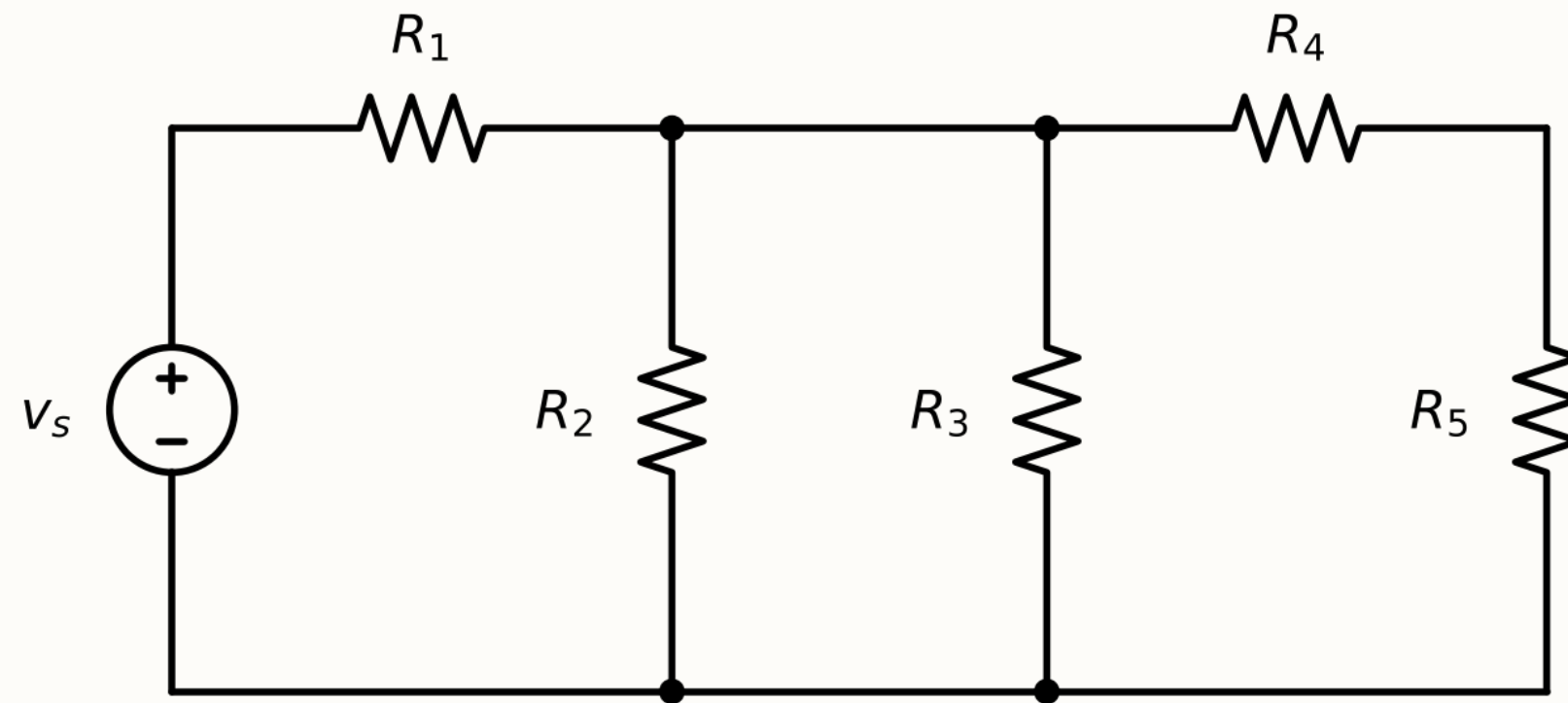
$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$$

two in parallel: $R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$

The equivalent is always smaller than the smallest resistor in the group.

PREDICT FIRST · 1 MINUTE

Which resistors are in series? Which are in parallel?



Look at the nodes, not the drawing.

Where does a third branch leave a node?

ANSWER

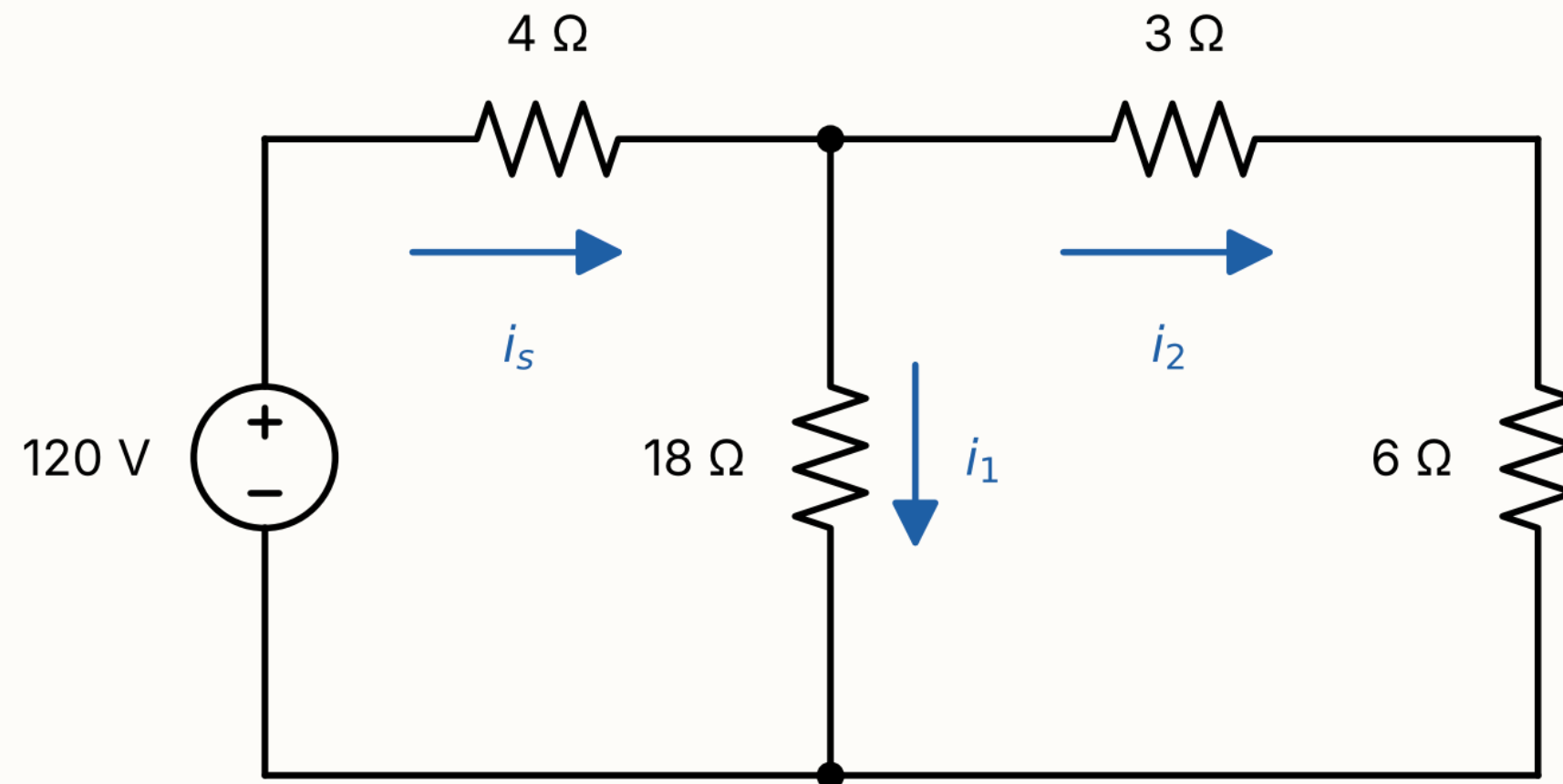
Two pairs; R_1 pairs with no single resistor

Pair	Relation	Why
R_2 and R_3	Parallel	Both connect the same two nodes
R_4 and R_5	Series	Only those two meet at the node between them
R_1 and R_2	Neither	R_3 and R_4 also leave the node between them

So the source sees $R_1 + R_2 \parallel R_3 \parallel (R_4 + R_5)$.

NILSSON EXAMPLE 3.2

Reduce from the far end, then work back



1. Combine away from the source:

$$3 + 6 = 9 \, \Omega, \quad 18 \parallel 9 = \frac{(18)(9)}{18 + 9} = 6 \, \Omega$$

2. One loop is left:

$$i_s = \frac{120}{4 + 6} = 12 \, \text{A}$$

NILSSON EXAMPLE 3.2

Expand back to find the branch currents, then check

v_1 is the voltage across the $18\ \Omega$, + at the top:

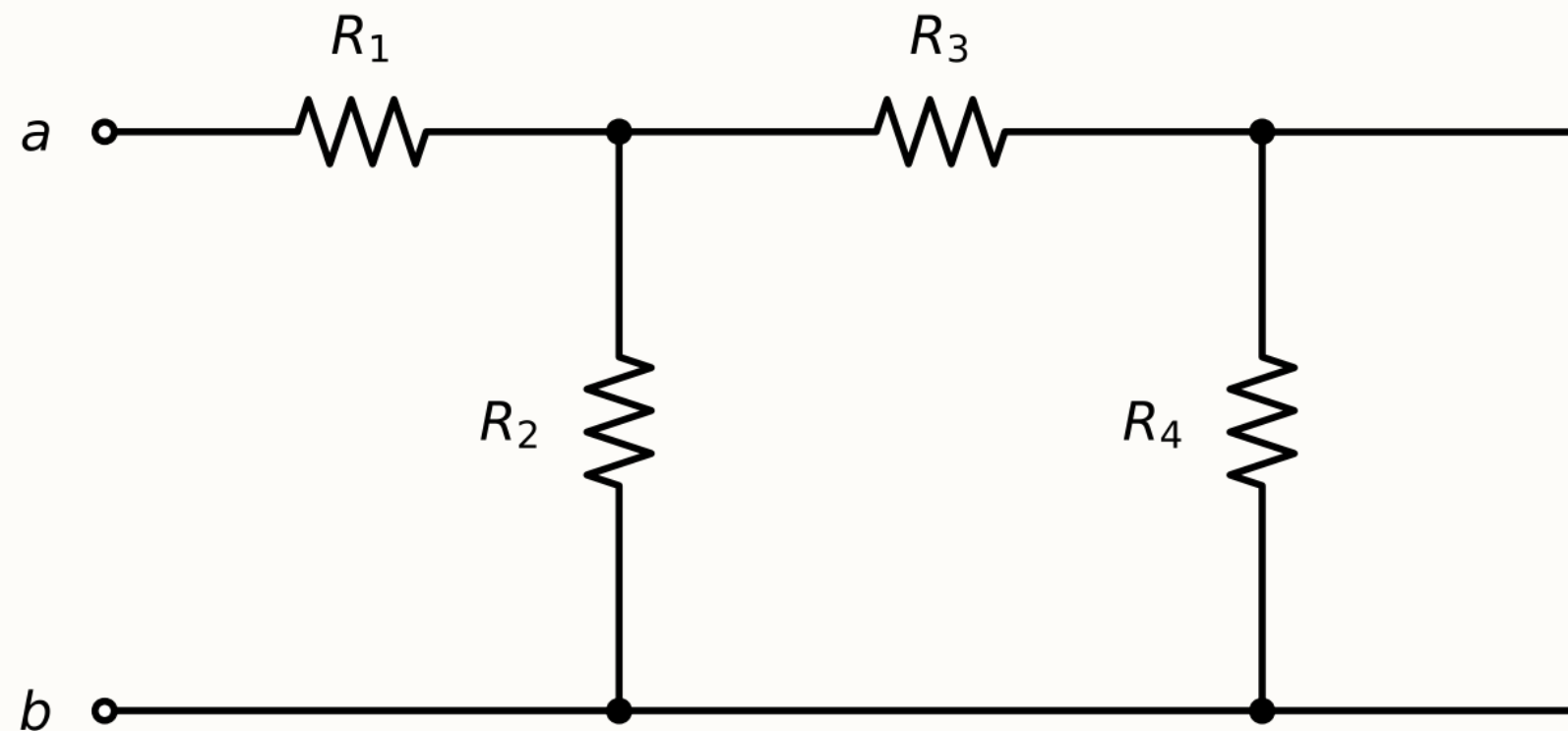
$$v_1 = (12)(6) = 72\ \text{V}, \qquad i_1 = \frac{72}{18} = 4\ \text{A}, \qquad i_2 = \frac{72}{9} = 8\ \text{A}$$

Element	Power	Absorbs or delivers
120 V source	−1440 W	delivers
4 Ω	576 W	absorbs
18 Ω	288 W	absorbs
3 Ω	192 W	absorbs
6 Ω	384 W	absorbs

Check: the $18\ \Omega$ and the $3\ \Omega + 6\ \Omega$ branch both have 72 V across them.

TWO TRAPS

A wire across a resistor, and a resistor in the bottom wire



A wire forces 0 V

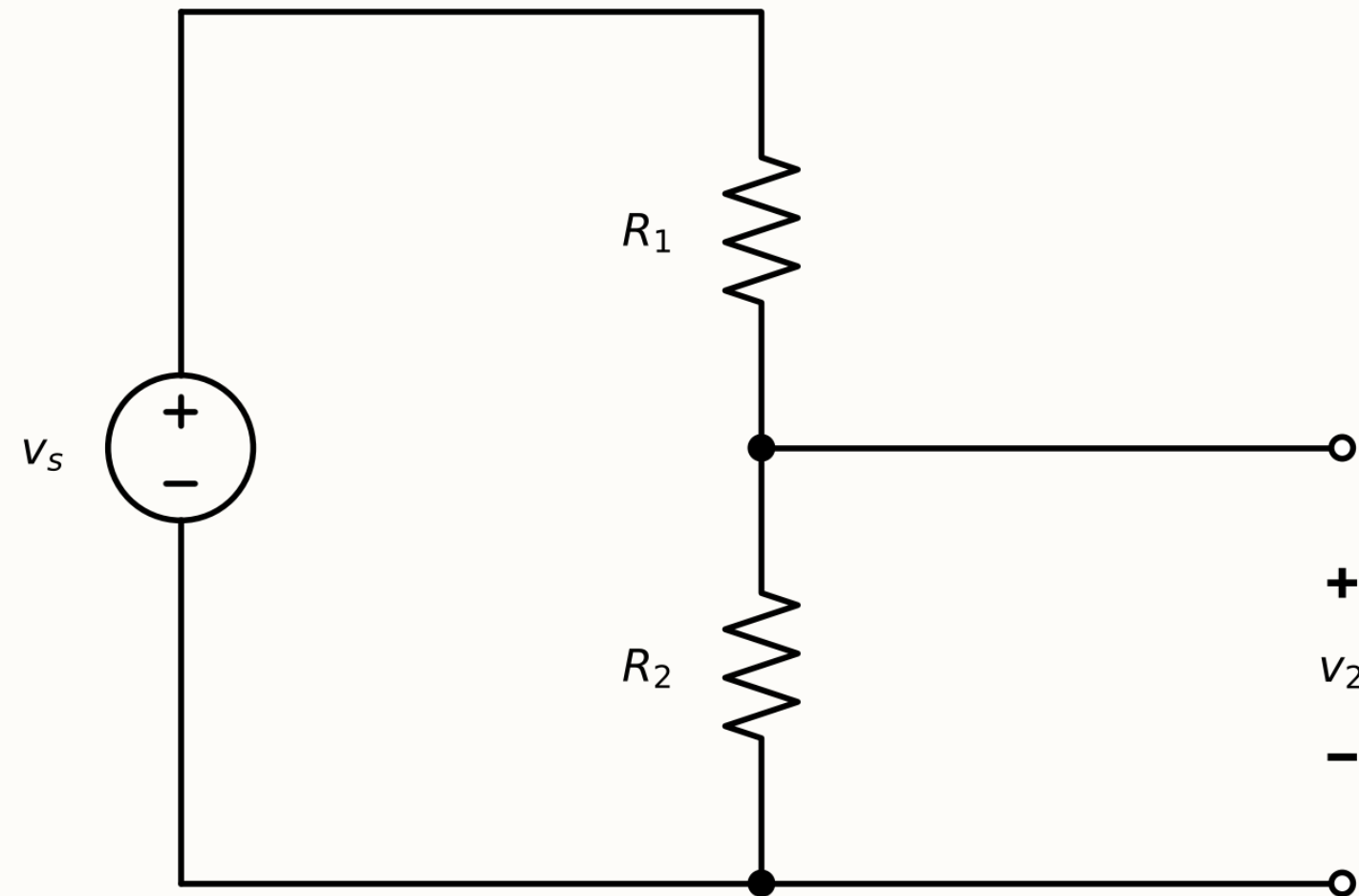
R_4 has 0 V across it, so it carries no current, and the resistance between a and b is $R_1 + R_2 \parallel R_3$.

The bottom wire counts

A resistor drawn in the return wire is as real as one on top. Booklet 3.3 and practice item S2 have one: include it.

VOLTAGE DIVISION

Series resistors share the voltage in proportion to their resistance



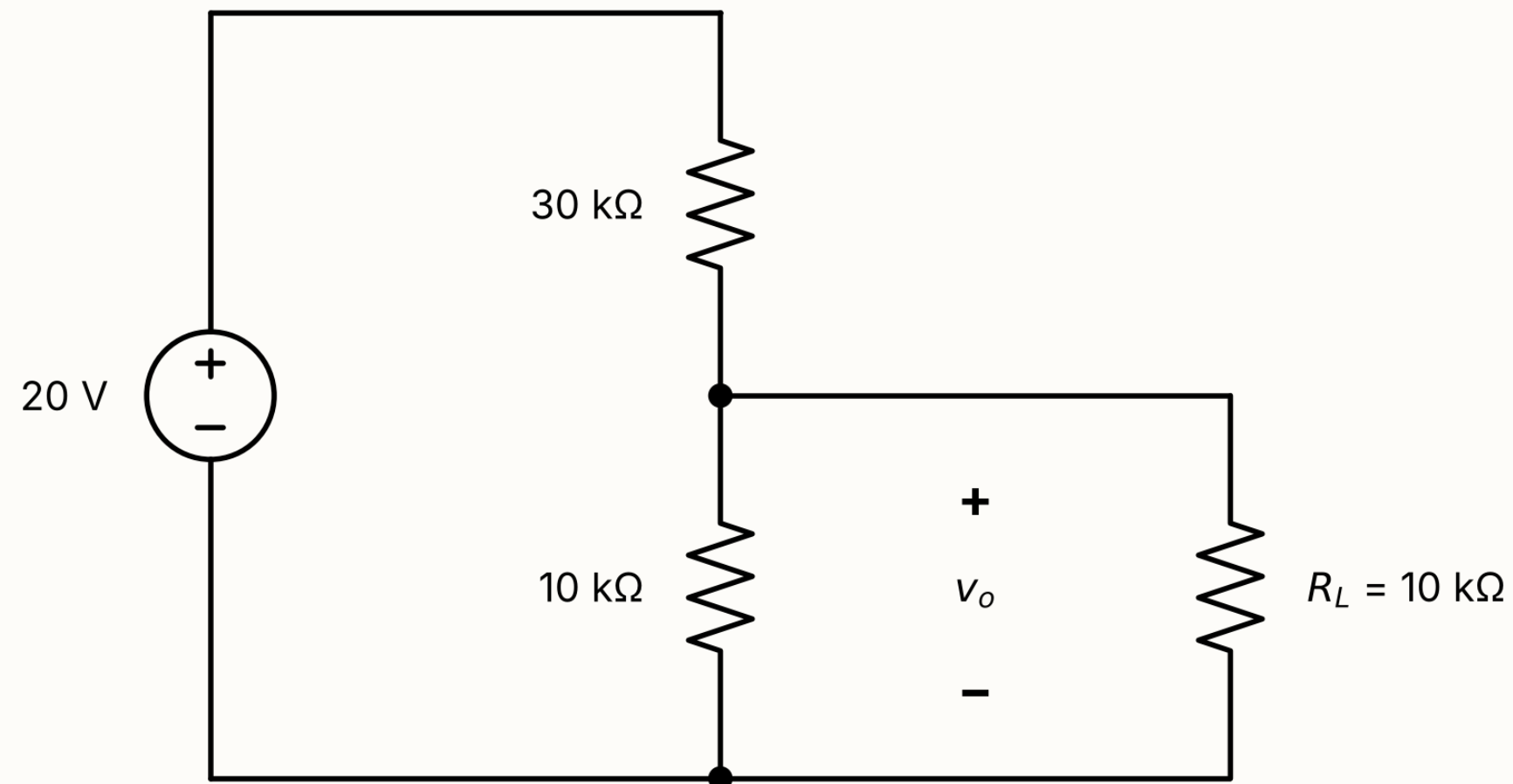
$$v_1 = \frac{R_1}{R_1 + R_2} v_s, \quad v_2 = \frac{R_2}{R_1 + R_2} v_s$$

in general: $v_j = \frac{R_j}{R_{eq}} v$

The share is always less than 1, so each part is smaller than the total.

NILSSON EXAMPLE 3.4

A load changes a voltage divider



Designed for 5 V. A 10 kΩ load in parallel with the 10 kΩ makes 5 kΩ:

$$v_o = \frac{5}{30 + 5}(20) = 2.86\text{ V}$$

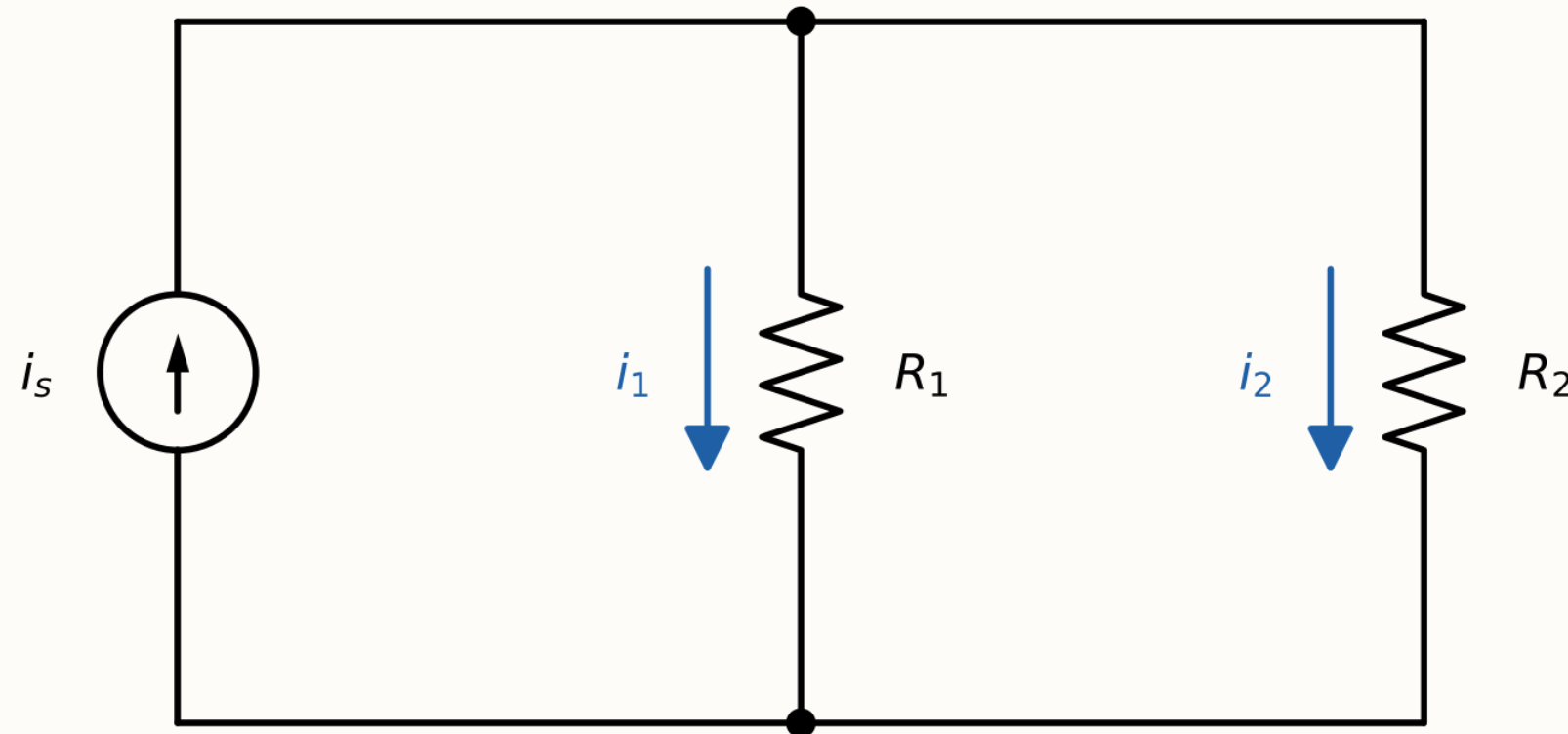
The same load on a 1200 Ω and 400 Ω divider makes 384.6 Ω:

$$v_o = \frac{384.6}{1200 + 384.6}(20) = 4.85\text{ V}$$

Rule: keep the load much larger than the bottom resistor.

CURRENT DIVISION

Parallel resistors share the current in inverse proportion

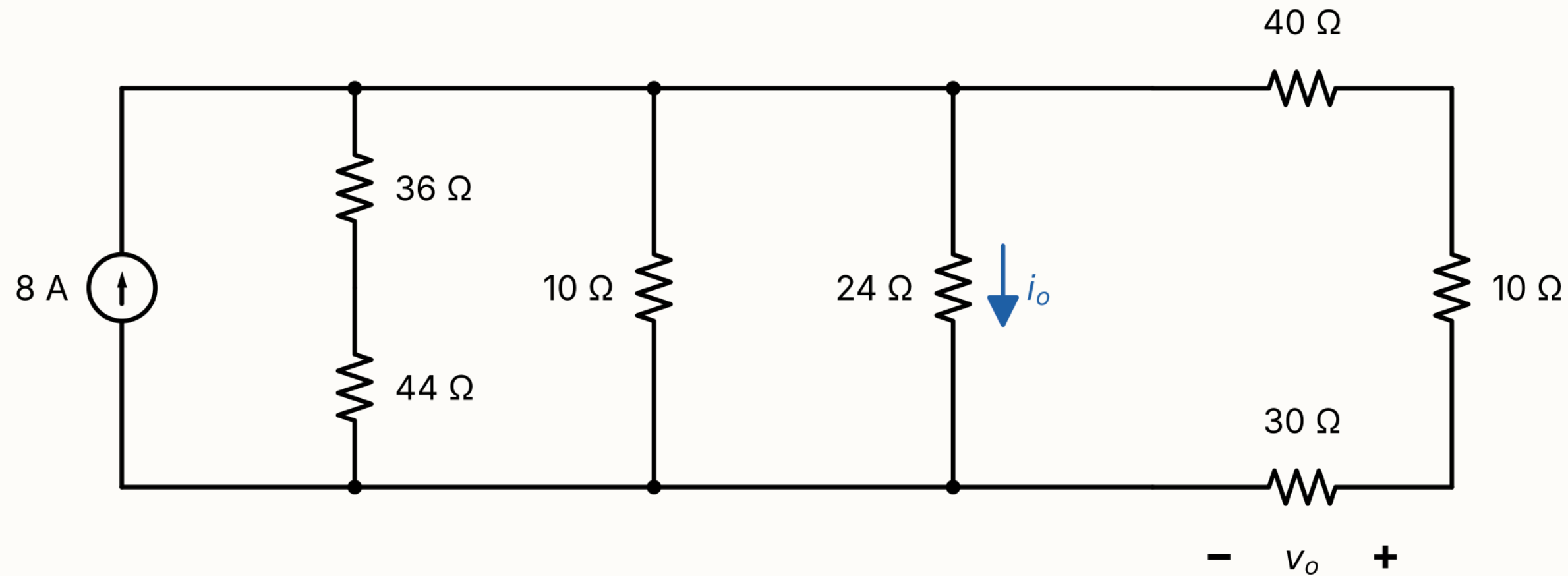


$$i_1 = \frac{R_2}{R_1 + R_2} i_s, \quad i_2 = \frac{R_1}{R_1 + R_2} i_s$$

in general: $i_j = \frac{R_{eq}}{R_j} i$

For two resistors, the **other** resistance goes on top: the smaller resistor takes the larger current.

Which branch takes the most current? Find i_o and v_o .



Current division gives i_o ; then voltage division in the $40 + 10 + 30 \Omega$ branch gives v_o .

NILSSON EXAMPLE 3.7

Divide the current, then divide the voltage

$$R_{eq} = (36 + 44) \parallel 10 \parallel (40 + 10 + 30) \parallel 24 = 80 \parallel 10 \parallel 80 \parallel 24 = 6 \Omega$$

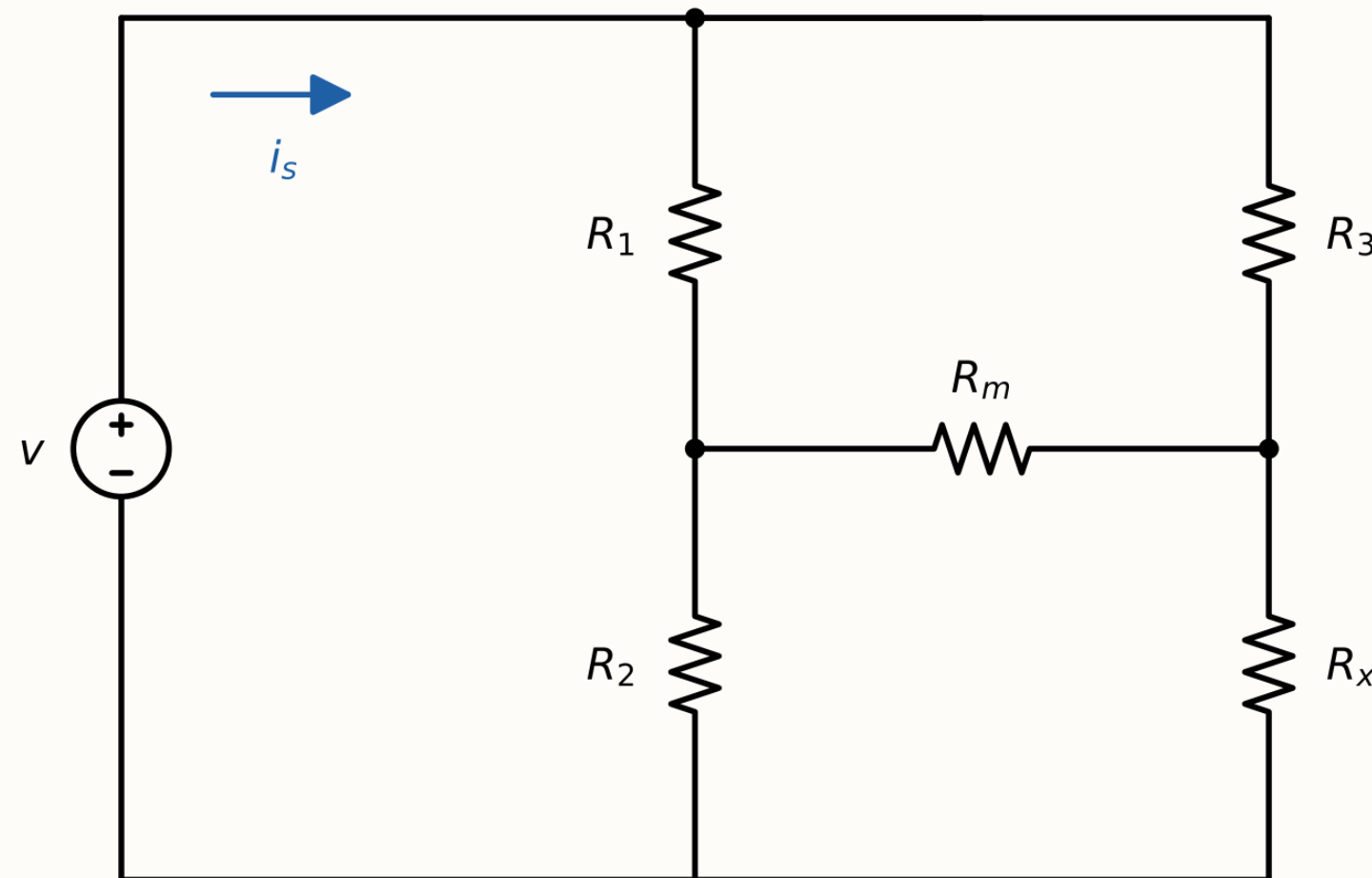
$$i_o = \frac{6}{24}(8) = 2 \text{ A}, \quad v = (24)(2) = 48 \text{ V}$$

$$v_o = \frac{30}{40 + 10 + 30}(48) = 18 \text{ V}$$

v is the voltage across the source, + at the top. The 10Ω branch takes the most current, 4.8 A; the four branch currents 0.6, 4.8, 2 and 0.6 A add to 8 A.

A BRIDGE

No two resistors here are in series or in parallel



Every node has three branches, so the series and parallel rules cannot start.

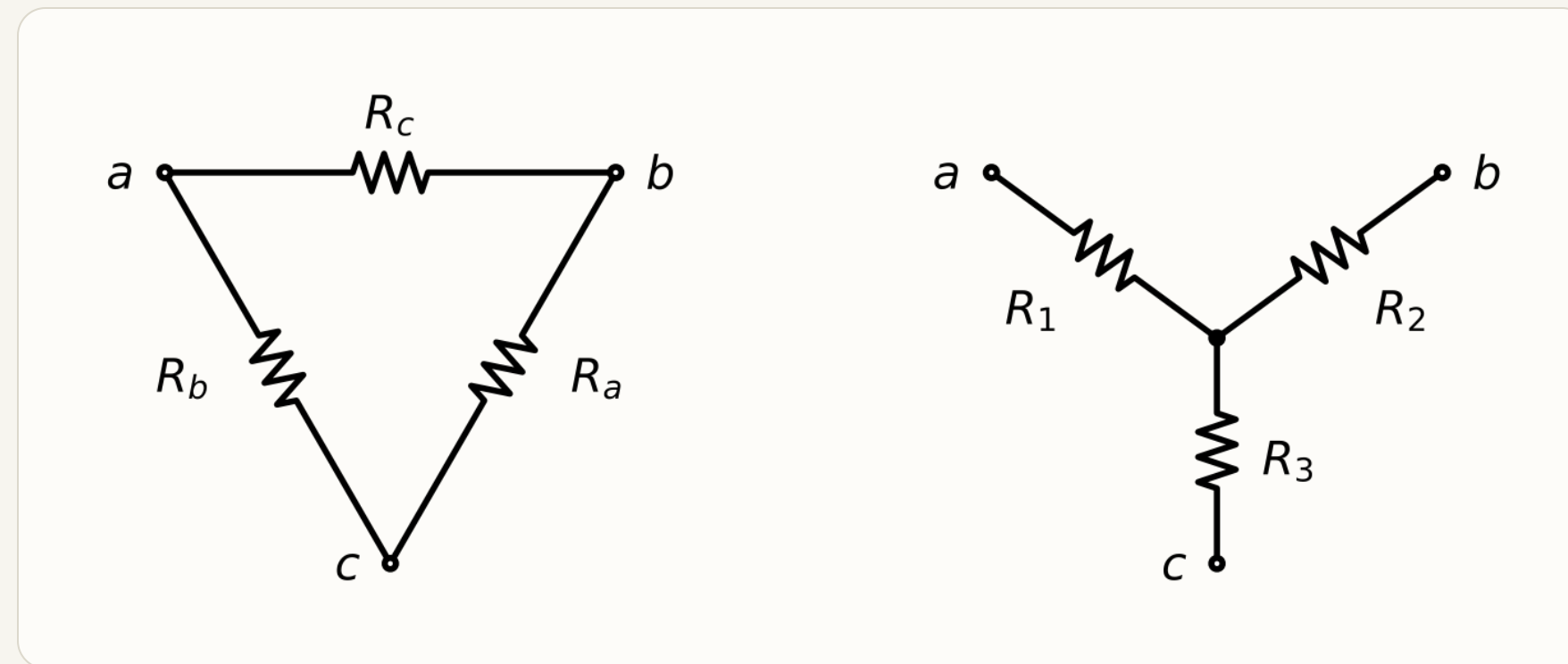
Two ways forward: a Δ -to-Y step (next slides), or the node-voltage method of Chapter 4.

balanced when $\frac{R_1}{R_2} = \frac{R_3}{R_x}$

Then R_m carries no current: the Wheatstone bridge.

DELTA TO WYE

Each Y resistor: the two Δ resistors at its terminal, over the sum



$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c}, \quad R_2 = \frac{R_c R_a}{R_a + R_b + R_c}, \quad R_3 = \frac{R_a R_b}{R_a + R_b + R_c}$$

The two circuits have the same resistance between every pair of terminals, so one can replace the other.

WYE TO DELTA

Each Δ resistor: the sum of products, over the opposite Y resistor

$$R_a = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1}, \quad R_b = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_2}, \quad R_c = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_3}$$

Which way?

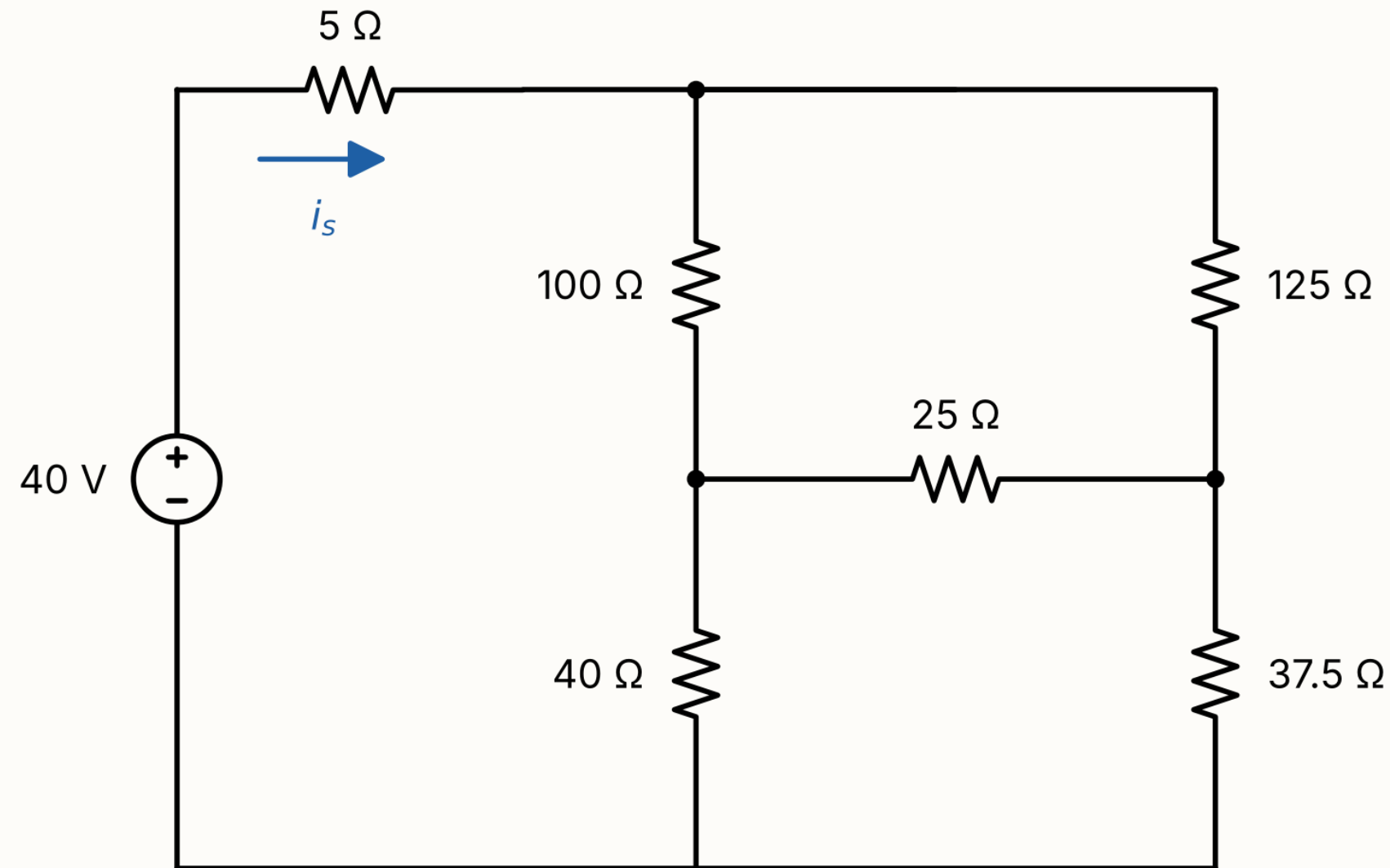
Δ to Y removes a loop of three resistors; Y to Δ removes a node. Pick the one that creates series or parallel pairs.

Check

Compute one terminal-pair resistance both ways. It must agree.

NILSSON EXAMPLE 3.11

Replace the upper Δ by a Y



The 100 Ω , 125 Ω and 25 Ω resistors form a Δ , with $\Sigma = 250 \Omega$.

$$R_1 = \frac{(100)(125)}{250} = 50 \Omega$$

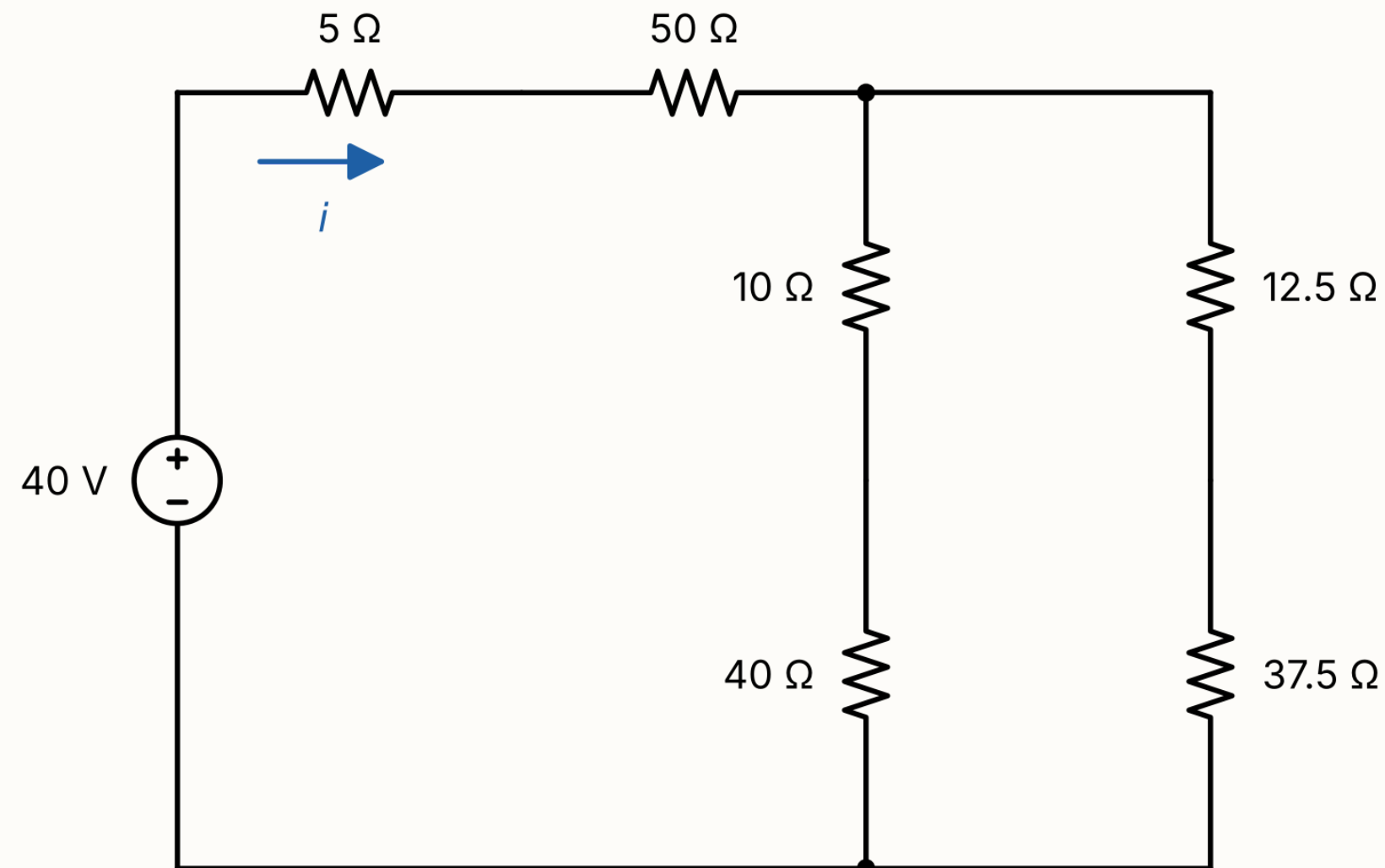
$$R_2 = \frac{(125)(25)}{250} = 12.5 \Omega$$

$$R_3 = \frac{(100)(25)}{250} = 10 \Omega$$

R_1 goes to the top node, R_2 to the right node, R_3 to the left node.

NILSSON EXAMPLE 3.11

Now series and parallel finish the job



$$R_{eq} = 5 + 50 + (10 + 40) \parallel (12.5 + 37.5) \\ = 55 + 25 = 80 \, \Omega$$

$$i = \frac{40}{80} = 0.5 \, \text{A}, \quad p = (40)(0.5) = 20 \, \text{W}$$

The 40 V source delivers 20 W.

CHECKING AN AI ANSWER

An AI found i_2 in Example 3.2 like this. What is wrong?

$$i_2 = \frac{9}{18 + 9}(12) = 4 \text{ A}$$

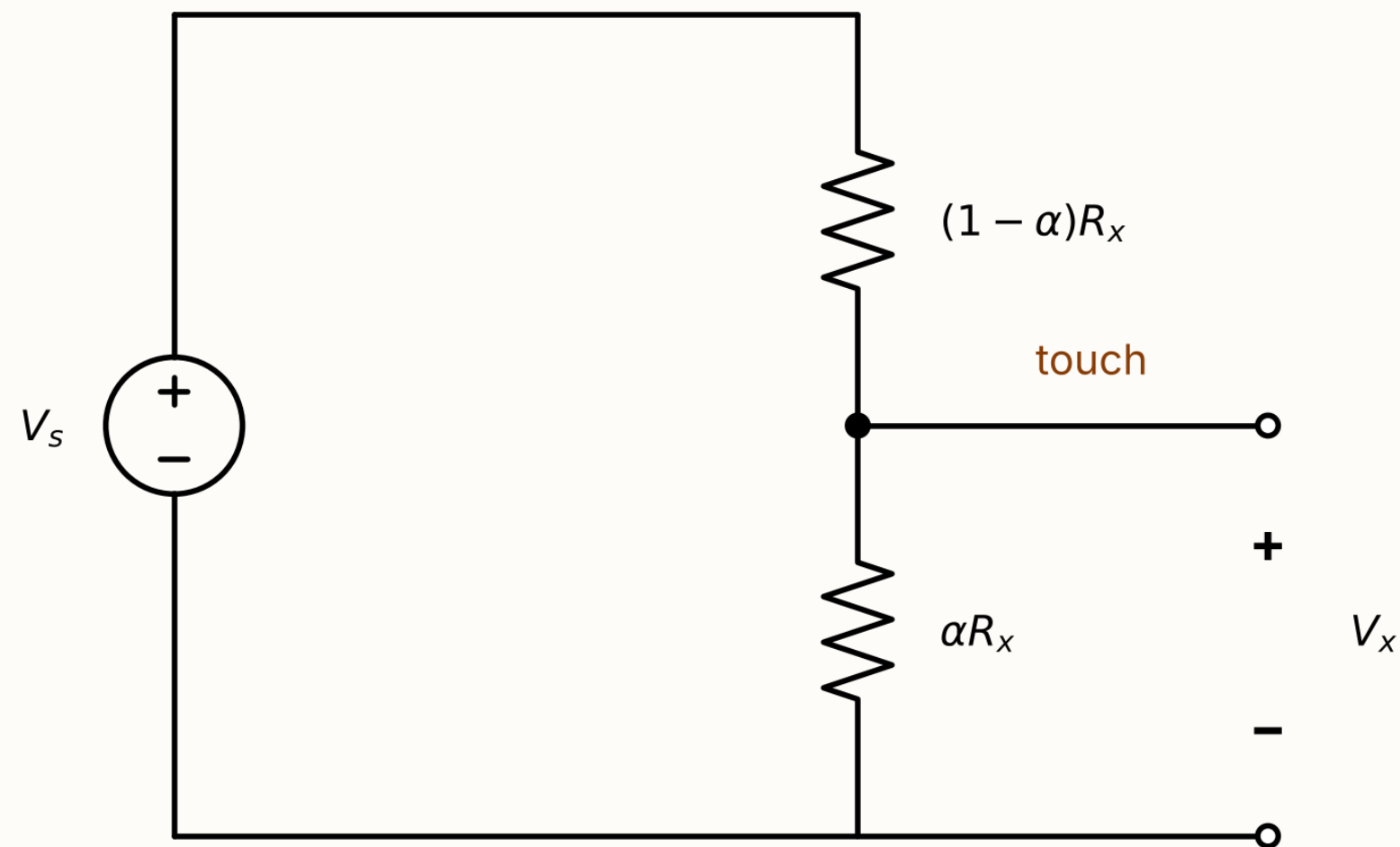
Correct:

$$i_2 = \frac{18}{18 + 9}(12) = 8 \text{ A}$$

The other branch's resistance goes on top. The giveaway: with 4 A in the 9 Ω branch and 8 A in the 18 Ω , the two parallel branches would have 36 V and 144 V.

In a 2025 study of Gemini 2.5 Pro on undergraduate circuit problems, misread source polarities caused about a third of its wrong answers, and misread current directions nearly as many.

One voltage divider per direction



$$V_x = \frac{\alpha R_x}{\alpha R_x + (1 - \alpha)R_x} V_s = \alpha V_s$$

$$x = (1 - \alpha) p_x$$

α is measured from the right edge; pixel columns are counted from the left. A screen 1080 pixels wide, $V_s = 5$ V:

$$V_x = 1 \text{ V} : \quad \alpha = \frac{1}{5} = 0.2$$

$$x = (0.8)(1080) = 864$$

YOUR SELF-CHECK

Three checks that need no device

Same voltage

Resistors in parallel must have the same voltage across them.

Never more than the total

A divided voltage or current can never exceed the total it came from.

Powers add to zero

$$\sum p = 0$$

Every source and every resistor.

WHAT COMES NEXT

Practise openly, then show it on your own

In class

Booklet Problems 3.1 to 3.10.
Problem 3.11 is homework.

At home

The 16-item practice sheet. AI
tools allowed; not marked.

Week 5

15-minute quiz, no devices.
Every question is a twin of a
practice-sheet question.

Chapter 3 is the last chapter on the midterm.

CHECKLIST

You can do these without help

- Say which resistors are in series and which are in parallel, and why
- Spot a wire that shorts out a resistor
- Find the equivalent resistance of a series-parallel network
- Use voltage division and current division, and check the result
- Explain why a load changes a divider's output
- Convert a Δ to a Y and a Y to a Δ , and use it to solve a bridge